Packaging Sustainability in Consumer Products

Introduction: Modern retail operations are experiencing mounting pressure to eliminate non-recyclable solid waste from global distribution chains.

Transitioning fast-moving consumer goods to circular packaging frameworks requires substituting virgin petroleum polymers with post-consumer recycled polyethylene terephthalate and industrially compostable polyhydroxyalkanoates. Achieving mechanical recyclability without sacrificing barrier properties against moisture and oxygen ingress represents a critical chemical engineering challenge. Successful circular supply loops require closed-loop reverse collection incentives, automated near-infrared optical flake sorting, and decontamination extrusion technologies to meet food-grade contact safety regulations.